

Do Health Management Chatbots Replace Our Health Decision-making?

Weiran Cui, Chi-Lan Yang, Takuji Narumi

Graduate School of Interdisciplinary Information Studies, The University of Tokyo

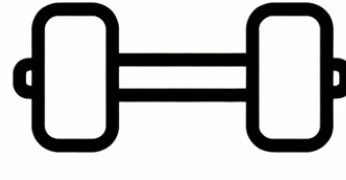
Introduction

Conversational AI systems are increasingly used to support everyday decision-making [1]. Prior research has raised concerns that increasing reliance on AI may reduce users' engagement in independent evaluation and reflection [2].

However, not all decision contexts are equivalent. Health-related decisions involve perceived personal risk and potential consequences for well-being [3]. Even in routine health management scenarios, choices may carry uncertainty or long-term impact. In such risk-sensitive domains, users may be more cautious and less willing to fully rely on AI-generated advice. Meanwhile, conversational agents are often designed and presented in role-like personas, such as a coach, counselor, peer/companion, or health professional [4]. These roles may influence how users interpret and evaluate advice, potentially affecting their willingness to rely on it. This study examines how different roles in health context shapes users' responses to health-related advice and discusses the implications for reliance and independent judgment.

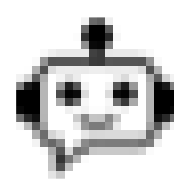
Method

4 role characteristics:

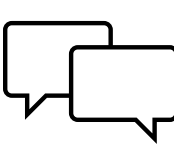
	Close	Distant
High expertise	 Coach (11): emphasizes expertise and encouragement	 Expert (13): emphasizes professional expertise, lower relational closeness
Low expertise	 Partner (11): emphasizes relational closeness	 Influencer (10): emphasizes personal experience

Procedures:

Task (25mins):

-  Self-introduction from the chatbot
-  Talk about health concerns
-  Set plans for a healthy lifestyle

Two-week interaction


Interview (20-30mins)

Participants:

- 45 (mean = 25.25; 24 female, 21 male)
- Randomly assigned: expert (13), coach (11), partner (11), influencer (10)

Qualitative Results

Perceived expertise was the primary basis of credibility in the health context. Participants evaluated expertise through:

- Logical, structured reasoning
- Clear step-by-step explanations
- Depth of domain knowledge

"I trust health advice more when it comes with a clear 'why.' If the logic behind the suggestion is sound, I'm much more likely to follow it. It just feels more reassuring that way." (Expert, P32)

Participants appreciated the sense of closeness and conversational warmth. This increased comfort and initial trust. However, closeness alone was insufficient to sustain reliance when in-depth or actionable guidance was lacking.

"I trust its accuracy, but I see it as a reference tool, not a guide. I rely on it for conceptual explanations, but I prefer to keep it out of my personal health management decisions." (Partner, P9)

Trust did not remain stable once established. Even when initial expertise cues were strong, participants continued to monitor responses against:

- Their prior knowledge
- Personal experience
- Occasionally external sources

"When deciding which advice to actually try, I usually combine its suggestions with my own past experiences to make a well-rounded judgment." (Expert, P40)
"If it's anything technical or professional, I'd probably double-check it on Google or other sources before I'd actually dare to put it into practice." (Coach, P24)

Discussion & Conclusion

Expertise was necessary for participants to take the advice seriously. While relational warmth made the interaction more comfortable, it did not compensate for a lack of perceived competence.

Even when participants initially trusted the chatbot, they did not follow its advice automatically. Instead, they checked whether the information matched their own knowledge and experience and sometimes verified it through other sources. Advice was more likely to be considered when it aligned with their expectations; when it did not, trust quickly declined. Overall, in health management contexts, conversational AI is primarily treated as a decision-support tool rather than a decision-maker. Even when roles increased initial trust, users retained independent judgment.

Acknowledgement

This work was supported by JST COI-NEXT (JPMJPF2201). We also thank all participants who took part in this study.

References

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Implications & Future Work

Health chatbots should help users think through decisions rather than make decisions for them. For example, systems can explain how they arrive at recommendations, point out possible limitations, and encourage users to double-check important information. This may strengthen trust while keeping users actively involved in the decision process.

Future research could examine whether patterns of reliance differ across domains with varying perceived risk (e.g., health decisions vs. shopping planning). Lower-stakes contexts may encourage increased dependence on AI suggestions.